

WAVEFORM LOGIC

Glossary of Mathematical Symbols, Operators, and Terms

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1. Spaces and Sets

Symbol	Name & Definition	Notes
$\mathbb{W}$	Waveform space. The set of all waveform numbers — the primary domain of Waveform Logic.	Not a field. A non-associative algebra with drift. Replaces $\mathbb{R}$ as the fundamental number space.
$\mathbb{R}$	Real numbers. The standard real number line.	The <i>image</i> of the collapse map $[\cdot]$. Every $r \in \mathbb{R}$ is the shadow of infinitely many $w \in \mathbb{W}$.
$\mathbb{R}^+$	Positive reals. The domain of the frequency parameter f .	Frequencies are strictly positive; zero frequency is undefined (a non-oscillating signal is not a waveform number).
$[0, 2\pi)$	Phase space. The domain of the phase parameter φ .	A circle: $\varphi = 0$ and $\varphi = 2\pi$ are identified. Phase wraps.

2. Primitive Parameters

Symbol	Name & Definition	Notes
f	Frequency. Cycles per unit time. Element of $\mathbb{R}^+$.	In the number mapping, $n \mapsto n \cdot f_0$ embeds the naturals into the harmonic series.
A	Amplitude. Maximum displacement from equilibrium. Element of $\mathbb{R}$.	Negative A is equivalent to positive A with π phase shift; the sign convention is kept for algebraic convenience.
φ	Phase. Angular offset at origin. Element of $[0, 2\pi)$.	Phase inversion ($\varphi \mapsto \varphi + \pi$) defines the additive inverse.
σ	Shape function. A periodic map $\sigma : \mathbb{R} \rightarrow [-1, 1]$ satisfying $\sigma(t + 1) = \sigma(t)$.	The fourth parameter and the source of most novelty. Sine, square, triangle, sawtooth, or any unit-periodic function.
f_0	Base frequency. The reference frequency corresponding to the number 1.	A freely chosen constant that anchors the number–frequency mapping. Analogous to choosing a unit of measure.

3. Objects

Symbol	Name & Definition	Notes
$w = \langle f, A, \varphi, \sigma \rangle$	Waveform number. The fundamental object of $\mathbb{W}$: a quadruple of frequency, amplitude, phase, and shape.	Two waveform numbers are equal iff all four parameters are identical. Same-value-at-an-instant does not imply equality.
$\hat{w}(t)$	Realisation. The instantaneous value of w at time t : $\hat{w}(t) = A \cdot \sigma(ft + \varphi)$.	Maps a waveform number to a time-varying real signal. The “playback” of the number.
e_0	Unit oscillator. The canonical reference waveform: $e_0 = \langle 1, 1, 0, \sin \rangle$.	Used as the probe signal in the collapse operation. The “measuring stick” of $\mathbb{W}$.
sq	Square wave. The fixed-point attractor of the drift operator in shape space: $\text{sq}(t) = \text{sgn}(\sin(2\pi t))$.	Every waveform converges to sq under repeated drift. The terminal shape of all oscillation.

4. Standard Operations (Wave Reinterpretations)

Symbol	Name & Definition	Notes
$w_1 + w_2$	Superposition. Pointwise sum of realisations: $\widehat{(w_1 + w_2)}(t) = \hat{w}_1(t) + \hat{w}_2(t)$.	Commutative, associative. Identity: zero-amplitude waveform. The wave analogue of addition.
$-w$	Phase inversion / negation. $\langle f, A, \varphi + \pi, \sigma \rangle$.	$w + (-w) = 0$. Destructive interference as additive inverse.
$w_1 - w_2$	Subtraction. Defined as $w_1 + (-w_2)$. Superposition after phase inversion.	Partial destructive interference between the two waveforms.
$\mathcal{S}(w)$	Spectrum. The set of all frequency components in a complex waveform number.	The Fourier decomposition reinterpreted as the identity of a number rather than a derived analysis.

5. Novel Operations (No Classical Analogue)

Symbol	Name & Definition	Notes
$w_1 \odot w_2$	Oscillation product. $\langle f_1 f_2, A_1 A_2, \varphi_1 + \varphi_2, \sigma_1 \otimes \sigma_2 \rangle$, where $(\sigma_1 \otimes \sigma_2)(t) = \sigma_1(\sigma_2(t) \cdot t)$.	Non-associative. Shape composition means the order of operands changes the result. Not ring multiplication. The core novel operation.
$\sigma_1 \otimes \sigma_2$	Shape composition. $(\sigma_1 \otimes \sigma_2)(t) = \sigma_1(\sigma_2(t) \cdot t)$. The shape function of σ_2 modulates the <i>argument</i> of σ_1 .	The engine of non-associativity. Produces waveform morphologies unreachable by FM, AM, or ring modulation.
$w \rightsquigarrow^\delta$	Drift. Simultaneous coupled transformation: $\langle f + \delta A, A \cdot e^{- \delta }, \varphi + \delta f, \sigma^{(\delta)} \rangle$ where $\sigma^{(\alpha)}(t) = (1 - \alpha)\sigma(t) + \alpha\sigma(t)^3$, clamped to $\alpha \in [0, 1]$.	No classical analogue. Shifts frequency, decays amplitude, rotates phase, and deforms shape—all coupled through a single parameter δ . Convergent: all waveforms drift toward sq.
$\gamma(w_1, w_2)$	Coherence. $\lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T \hat{w}_1(t) \hat{w}_2(t) dt$.	Shape-sensitive scalar pairing. Unlike an inner product, $\gamma(w, w) \neq A^2$ for non-sinusoidal σ . Defines a non-Euclidean geometry on $\mathbb{W}$.
$w_1 \bowtie w_2$	Entanglement. Structural dependency: $\exists \Phi : \mathbb{W} \rightarrow \mathbb{W}$ such that any operation T on w_1 induces $\Phi \circ T$ on w_2 .	Not quantum entanglement. A purely algebraic constraint. Entangled numbers cannot be manipulated independently. Symmetric, not transitive.
$\lfloor w \rfloor$	Collapse. Projection $\mathbb{W} \rightarrow \mathbb{R}$ defined by $\lfloor w \rfloor = A \cdot \gamma(w, e_0)$.	Irreversible. Many-to-one. Real arithmetic operates on collapsed shadows of the richer $\mathbb{W}$ structure.

6. Derived Operations

Symbol	Name & Definition	Notes
$w^{\odot n}$	Harmonic stacking / exponentiation. The n -fold oscillation product of w with itself: $w \odot w \odot \dots \odot w$.	Because $\odot$ is non-associative, this must be evaluated left-to-right by convention. Different parenthesisations yield different results.
$\log_{\odot}(w_1, w_2)$	Harmonic order / logarithm. The value n such that $w_1^{\odot n} = w_2$, if it exists.	May not exist or may not be unique, due to non-associativity. Well-defined only for waveform numbers in harmonic relationship.
$\sigma^{(\alpha)}$	Shape deformation. $(1 - \alpha)\sigma(t) + \alpha\sigma(t)^3$, where $\alpha \in [0, 1]$.	The cubing term “squares off” any smooth shape. At $\alpha = 1$, most shapes closely approximate the square wave.
$d_\gamma(w_1, w_2)$	Coherence distance. A distance metric derived from coherence: $d_\gamma = 1 - \frac{ \gamma(w_1, w_2) }{\sqrt{\gamma(w_1, w_1) \cdot \gamma(w_2, w_2)}}$	Defines the geometry of $\mathbb{W}$. Shape-sensitive: two waveforms with equal f, A, φ but different σ are at non-zero distance.

7. Axiom Reference

Axiom	Statement
A1	Every element of $\mathbb{W}$ is a quadruple $\langle f, A, \varphi, \sigma \rangle$ with $f \in \mathbb{R}^+$, $A \in \mathbb{R}$, $\varphi \in [0, 2\pi)$, $\sigma : \mathbb{R} \rightarrow [-1, 1]$ unit-periodic.
A2	Superposition $(+)$ is commutative and associative, with identity $\langle f, 0, 0, \sigma \rangle$ for any f, σ .
A3	Phase inversion ($\varphi \mapsto \varphi + \pi$) yields the additive inverse: $w + (-w) = 0$.
A4	The oscillation product $\odot$ is commutative but <i>not</i> associative.
A5	$\odot$ distributes over $+$ from the left: $w_1 \odot (w_2 + w_3) = (w_1 \odot w_2) + (w_1 \odot w_3)$.
A6	Drift is a one-parameter semigroup: $w^{\rightsquigarrow 0} = w$; $(w^{\rightsquigarrow \alpha})^{\rightsquigarrow \beta} = w^{\rightsquigarrow (\alpha + \beta)}$ up to shape deformation.
A7	Entanglement ($\bowtie$) is symmetric but not transitive.
A8	Collapse $[\cdot]$ is a surjective homomorphism from $(\mathbb{W}, +)$ to $(\mathbb{R}, +)$.
A9	No element of $\mathbb{W}$ is fully determined by its collapse. For every $r \in \mathbb{R}$, there exist infinitely many $w \in \mathbb{W}$ with $[w] = r$.

8. Key Terms (Alphabetical)

Term	Definition
Carrier	A waveform that serves as the substrate for modulation by another signal. In $\mathbb{W}$, the first operand of $\odot$.
Coherence distance	The metric d_γ on $\mathbb{W}$ derived from the coherence measure. Shape-sensitive and non-Euclidean.
Coherence geometry	The geometry of $\mathbb{W}$ induced by the coherence measure γ . Distance depends on waveform morphology.
Collapse	The irreversible, many-to-one projection $[\cdot] : \mathbb{W} \rightarrow \mathbb{R}$. Conventional real numbers are collapsed waveform numbers.
Constructive interference	Superposition of in-phase waveforms producing increased amplitude. The wave-analogue of addition of like-signed numbers.
Convergence (drift)	The property that repeated drift drives any waveform toward the square wave fixed point sq.
Destructive interference	Superposition of anti-phase waveforms producing decreased amplitude. The mechanism underlying subtraction and zero.
Drift	The operator $(\rightsquigarrow)$ that simultaneously deforms all four parameters of a waveform number through a single scalar δ .
Entangled pair	Two waveform numbers $w_1 \bowtie w_2$ linked by a constraint function Φ such that operations on one propagate to the other.
Envelope	The function describing amplitude variation over time for a wave packet. In $\mathbb{W}$, tracked through the amplitude parameter A .
Filter	An operation that selectively attenuates frequency components. The wave-analogue of division.
Fixed-point attractor	The square wave sq, toward which all shapes converge under drift. The terminal morphology.
Harmonic	A frequency component that is an integer multiple of a fundamental frequency f_0 . The natural numbers as harmonics.
Interference	The combination of two or more waveforms producing a resultant. Serves as the definition of addition in $\mathbb{W}$.
Irreducible pair	An entangled pair for which no decomposition into independent components exists. The fundamental unit of relational arithmetic.
Non-associative algebra	An algebraic structure in which $(a \cdot b) \cdot c \neq a \cdot (b \cdot c)$ in general. $(\mathbb{W}, +, \odot)$ is such a structure.
Oscillation	A single complete cycle of a wave. The primitive “event” of Waveform Logic.
Oscillation product	The operation $\odot$ that combines two waveform numbers through shape composition. Non-associative by construction.
Realisation	The time-domain signal $\hat{w}(t)$ obtained by “playing” a waveform number. Maps $\mathbb{W}$ to the space of real-valued functions.
Resonance	The condition of shared frequency or harmonic relationship between waveforms. Produces amplitude amplification.
Shape composition	The operation $\otimes$ in which one shape function modulates the argument of another: $(\sigma_1 \otimes \sigma_2)(t) = \sigma_1(\sigma_2(t) \cdot t)$.
Shape deformation	The continuous transformation $\sigma^{(\alpha)}$ that blends a shape toward its own cube, driving convergence to sq.
Shape function	The periodic function $\sigma : \mathbb{R} \rightarrow [-1, 1]$ that defines the “character” of a waveform number beyond its frequency, amplitude, and phase.
Spectrum	The complete set of frequency components in a complex waveform number. The Fourier-domain identity of a number.
Superposition	The pointwise sum of waveform realisations. The definition of addition in $\mathbb{W}$. Linear, commutative, associative.
Unit oscillator	The canonical reference $e_0 = \langle 1, 1, 0, \sin \rangle$. The “1” of Waveform Logic and the probe for collapse.
Waveform number	The fundamental object of $\mathbb{W}$: a quadruple $\langle f, A, \varphi, \sigma \rangle$ encoding frequency, amplitude, phase, and shape.
Waveform space	See $\mathbb{W}$.

*This glossary accompanies the Waveform Logic formal specification (v0.1).
All definitions are working proposals subject to revision.*